Pearson Edexcel A-Level Mathematics (9MA0)

# Module 1

## Core Foundation & Coordinate Geometry

Paper 1 & 2 (Pure) · 87 practice questions · answers and mark schemes at the back

| Subtopic | Questions | Done |
|---|---|---|
| 1.1 Algebraic Expressions | 18 | ____ / 18 |
| 1.2 Quadratics | 18 | ____ / 18 |
| 1.3 Equations and Inequalities | 18 | ____ / 18 |
| 1.4 Straight Line Graphs | 15 | ____ / 15 |
| 1.5 Circles | 18 | ____ / 18 |

### How to use this workbook

1. Read the key facts and watch one video for the subtopic. Tick the skills checklist as you go.
2. Attempt each question in the squared working space without notes. Mark it with the scheme at the back.
3. Tick **Todo** when you have attempted it and **Mastered** only when you can redo it cold a week later.
4. Copy any mistakes into the notes box at the end of the subtopic. Rebuild the workbook with a new seed for fresh questions.

Name _____ Started _____ Built 14 Oct 2026, seed 2026

# 1.1 Algebraic Expressions

Manipulating powers, brackets and surds is the grammar of every later topic. Most marks are lost to sign slips and to not writing answers in the exact form asked for.

## Key facts

- $a^m \times a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$
- $a^{-n} = \frac{1}{a^n}$, $a^{\frac{1}{n}} = \sqrt[n]{a}$, $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$, $a^0 = 1$
- $\sqrt{ab} = \sqrt{a}\sqrt{b}$, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$
- Rationalise $\frac{1}{a+\sqrt{b}}$ by multiplying top and bottom by $a - \sqrt{b}$
- $a^2 - b^2 = (a-b)(a+b)$

## Method

- Write every term as a power of the same base before combining.
- Turn roots into fractional powers before differentiating or integrating.
- Factorise fully: take out the HCF first, then look for a quadratic or difference of two squares.
- For surds, find the largest square factor: $\sqrt{72} = \sqrt{36}\sqrt{2} = 6\sqrt{2}$.

## Exam traps

- $(3x)^2 = 9x^2$, not $3x^2$.
- $\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$.
- $\frac{1}{\sqrt{x}} = x^{-\frac{1}{2}}$ (the minus sign is the most dropped mark in Paper 1).
- 'Factorise fully' means no common factor left inside any bracket.

**In the exam:** Appears as quick 2-4 mark openers on Paper 1 and Paper 2, and inside almost every calculus question.

## Specification checklist

| Skill | Learnt | Practised | Mastered |
|---|---|---|---|
| Index laws | ☐ | ☐ | ☐ |
| Expanding brackets | ☐ | ☐ | ☐ |
| Factorising | ☐ | ☐ | ☐ |
| Negative and fractional indices | ☐ | ☐ | ☐ |
| Surds | ☐ | ☐ | ☐ |
| Rationalising denominators | ☐ | ☐ | ☐ |

## Videos (click to open)

☐ **Recap** — Algebraic Expressions in less than 20 minutes • A-Level Maths, Pure Year 1, Chapter 1 (Bicen Maths)
☐ Index laws — Indices: The Laws of Indices (TLMaths)
☐ Expanding brackets — Polynomials: Expanding Three or More Brackets (TLMaths)
☐ Factorising — Quadratics: Factorising Quadratics of the form ax^2 + bx + c (TLMaths)
☐ Negative and fractional indices — Indices: Examples of Negative Indices (TLMaths)
☐ Surds — Surds: Introducing Surds and Simplifying Surds (TLMaths)
☐ Rationalising denominators — Surds: Introducing Rationalising the Denominator Part 1 (TLMaths)

**Q1.1.01** Expanding brackets ●○○ [3 marks] ☐ Todo ☐ Mastered

Expand and simplify $(x+2)(x-1)(2x-5)$.

**(3)**

WORKING

**Q1.1.02** Surds ●○○ [2 marks] ☐ Todo ☐ Mastered

Simplify $\sqrt{96}-\sqrt{24}$, giving your answer in the form $k\sqrt{6}$.

**(2)**

WORKING

**Q1.1.03** Negative and fractional indices ●○○ [2 marks] ☐ Todo ☐ Mastered

Without using a calculator, find the exact value of $27^{\frac{1}{3}}$.

**(2)**

WORKING

**Q1.1.04** Factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Factorise fully $2x^3+6x^2-56x$.

**(3)**

WORKING

**Q1.1.05** Expanding brackets ●○○ [3 marks] ☐ Todo ☐ Mastered

Expand and simplify $(x+2)(x-5)(x+4)$.

**(3)**

WORKING

**Q1.1.06** Surds ●○○ [2 marks] ☐ Todo ☐ Mastered

Simplify $\sqrt{54} - \sqrt{216}$, giving your answer in the form $k\sqrt{6}$.

**(2)**

WORKING

**Q1.1.07** Negative and fractional indices ●○○ [2 marks] ☐ Todo ☐ Mastered

Without using a calculator, find the exact value of $49^{-\frac{1}{2}}$.

**(2)**

WORKING

**Q1.1.08** Factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Factorise fully $27x^3 - 75x$.

**(3)**

WORKING

**Q1.1.09** Expanding brackets ●○○ [3 marks] ☐ Todo ☐ Mastered

Expand and simplify $(x + 6)(x - 1)(x + 2)$.

**(3)**

WORKING

**Q1.1.10** **Surds** ●○○ **[2 marks]** ☐ Todo ☐ Mastered

Simplify $\sqrt{27} + \sqrt{75}$, giving your answer in the form $k\sqrt{3}$.

**(2)**

WORKING

**Q1.1.11** **Negative and fractional indices** ●○○ **[2 marks]** ☐ Todo ☐ Mastered

Without using a calculator, find the exact value of $16^{\frac{1}{2}}$.

**(2)**

WORKING

**Q1.1.12** **Index laws** ●○○ **[2 marks]** ☐ Todo ☐ Mastered

Simplify fully $\frac{(4x^4)^3}{3x^5}$.

**(2)**

WORKING

**Q1.1.13** **Rationalising denominators** ●○○ **[3 marks]** ☐ Todo ☐ Mastered

Express $\frac{2}{5+\sqrt{6}}$ in the form $a + b\sqrt{6}$, where $a$ and $b$ are rational.

**(3)**

WORKING

**Q1.1.14** Factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Factorise fully $x^3 + 7x^2 + 6x$.

**(3)**

WORKING

**Q1.1.15** Index laws ●●○ [3 marks] ☐ Todo ☐ Mastered

Solve $9^x \times 3^{x+4} = 9^{-2}$.

**(3)**

WORKING

**Q1.1.16** Rationalising denominators ●●○ [4 marks] ☐ Todo ☐ Mastered

Show that $\frac{5+4\sqrt{7}}{3+3\sqrt{7}}$ can be written as $p + q\sqrt{7}$, where $p$ and $q$ are rational, and find $p$ and $q$.

**(4)**

WORKING

**Q1.1.17** Index laws ●●○ [3 marks] ☐ Todo ☐ Mastered

Solve $8^x \times 2^{x+1} = 4^3$.

**(3)**

WORKING

**Q1.1.18** Rationalising denominators ●●○ [4 marks] ☐ Todo ☐ Mastered

Show that $\frac{1+1\sqrt{5}}{4+3\sqrt{5}}$ can be written as $p+q\sqrt{5}$, where $p$ and $q$ are rational, and find $p$ and $q$.

**(4)**

WORKING

MY NOTES — 1.1 ALGEBRAIC EXPRESSIONS: MISTAKES TO REMEMBER

# 1.2 Quadratics

Quadratics are solved by factorising, completing the square or the formula. The discriminant tells you how many roots there are and powers many 'find k' questions.

## Key facts

- $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4} + c$
- Turning point of $y = a(x + p)^2 + q$ is $(-p, q)$
- $b^2 - 4ac > 0$: two distinct roots; $= 0$: one repeated root; $< 0$: no real roots
- Line $y = mx + c$ is a tangent to a curve when the resulting quadratic has $b^2 - 4ac = 0$

## Method

- To complete the square with $a \neq 1$, take $a$ out of the $x^2$ and $x$ terms first.
- For 'find the range of k', write the discriminant condition, then solve the resulting inequality (often itself a quadratic: sketch it!).
- Hidden quadratics: substitute $u = x^2$, $u = \sqrt{x}$, $u = e^x$ or $u = 2^x$ and solve for $u$ first.
- Sketching: roots, $y$-intercept, turning point, and the shape from the sign of $a$.

## Exam traps

- Forgetting the 'show that' needs every line of algebra.
- Not rejecting impossible values of $u$ (e.g. $e^x = -2$).
- Mixing up $>$ and $\geq$ for 'real roots' (which includes equal roots: $\geq 0$).

**In the exam:** Discriminant 'find k' questions are 4-5 marks; completing the square often leads into a graph or a transformation question.

## Specification checklist

| Skill | Learnt | Practised | Mastered |
|---|---|---|---|
| Solving by factorising | ☐ | ☐ | ☐ |
| Quadratic formula | ☐ | ☐ | ☐ |
| Completing the square | ☐ | ☐ | ☐ |
| The discriminant | ☐ | ☐ | ☐ |
| Quadratic functions and graphs | ☐ | ☐ | ☐ |
| Hidden quadratics | ☐ | ☐ | ☐ |

## Videos (click to open)

☐ **Recap** — Quadratics in less than 28 minutes • A-Level Maths, Pure Year 1, Chapter 2 (Bicen Maths)
☐ Solving by factorising — Quadratics: Factorising Quadratics using the Difference of Two Squares (TLMaths)
☐ Quadratic formula — Quadratics: Where the Quadratic Formula Comes From (TLMaths)
☐ Completing the square — Quadratics: Examples of Completing the Square with the form ax^2 + bx + c (TLMaths)
☐ The discriminant — Quadratics: Using the Discriminant to Find How Many Roots a Quadratic Has (TLMaths)
☐ Quadratic functions and graphs — Graphs: Sketching Quadratics, Cubics, Quartics and Quintics (TLMaths)
☐ Hidden quadratics — Exponential Equations: Spotting Hidden Quadratics (TLMaths)

**Q1.2.01** Quadratic functions and graphs ●○○ [6 marks] ☐ Todo ☐ Mastered

The curve $C$ has equation $y = x^2 - 9$.

(a) Find the coordinates of the points where $C$ meets the coordinate axes. **(3)**

(b) Find the coordinates of the turning point of $C$, stating whether it is a maximum or a minimum. **(3)**

WORKING

**Q1.2.02** Quadratic formula ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $2x^2 - 2x - 9 = 0$, giving your answers in exact simplified surd form.

**(3)**

WORKING

**Q1.2.03** Solving by factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $2x^2 - 4x - 30 = 0$.

**(3)**

WORKING

**Q1.2.04** Quadratic functions and graphs ●○○ [6 marks] ☐ Todo ☐ Mastered

The curve $C$ has equation $y = 25 - x^2$.

(a) Find the coordinates of the points where $C$ meets the coordinate axes. **(3)**

(b) Find the coordinates of the turning point of $C$, stating whether it is a maximum or a minimum. **(3)**

WORKING

**Q1.2.05** Quadratic formula ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $x^2 - 2x - 7 = 0$, giving your answers in exact simplified surd form.

**(3)**

WORKING

**Q1.2.06** Solving by factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $2x^2 + 6x + 4 = 0$.

**(3)**

WORKING

**Q1.2.07** Completing the square ●○○ [3 marks] ☐ Todo ☐ Mastered

(a) Write $x^2 - x$ in the form $(x + p)^2 + q$, where $p$ and $q$ are constants. **(2)**

(b) Hence state the coordinates of the turning point of $y = x^2 - x$. **(1)**

WORKING

**Q1.2.08** Quadratic functions and graphs ●○○ [6 marks] ☐ Todo ☐ Mastered

The curve $C$ has equation $y = -x^2 + 9x - 20$.

(a) Find the coordinates of the points where $C$ meets the coordinate axes. **(3)**

(b) Find the coordinates of the turning point of $C$, stating whether it is a maximum or a minimum. **(3)**

WORKING

**Q1.2.09** Quadratic formula ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $3x^2 + 6x - 7 = 0$, giving your answers in exact simplified surd form.

**(3)**

WORKING

**Q1.2.10** Solving by factorising ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve $x^2 + 3x - 10 = 0$.

**(3)**

WORKING

**Q1.2.11** Completing the square ●●○ [4 marks] ☐ Todo ☐ Mastered

Express $-2x^2 - 4x + 1$ in the form $a(x + b)^2 + c$, where $a$, $b$ and $c$ are integers. Hence state the maximum value of $-2x^2 - 4x + 1$.

**(4)**

WORKING

**Q1.2.12** **The discriminant** ●●○ **[3 marks]** ☐ Todo ☐ Mastered

The equation $x^2 + 3x + k = 0$, where $k$ is a constant, has no real roots. Find the range of possible values of $k$.

**(3)**

WORKING

**Q1.2.13** **Hidden quadratics** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Solve $x - 5\sqrt{x} + 4 = 0$.

**(4)**

WORKING

**Q1.2.14** **Completing the square** ●●○ **[4 marks]** ☐ Todo ☐ Mastered

Express $-x^2 - 6x - 9$ in the form $a(x + b)^2 + c$, where $a$, $b$ and $c$ are integers. Hence state the maximum value of $-x^2 - 6x - 9$.

**(4)**

WORKING

**Q1.2.15** The discriminant ●●○ [3 marks] ☐ Todo ☐ Mastered

The equation $2x^2 - 6x + k = 0$, where $k$ is a constant, has two distinct real roots. Find the range of possible values of $k$.

**(3)**

WORKING

**Q1.2.16** Hidden quadratics ●●○ [4 marks] ☐ Todo ☐ Mastered

Find all real solutions of $x^4 - 5x^2 + 4 = 0$.

**(4)**

WORKING

**Q1.2.17** Hidden quadratics ●●○ [4 marks] ☐ Todo ☐ Mastered

Find all real solutions of $x^4 - 17x^2 + 16 = 0$.

**(4)**

WORKING

**Q1.2.18** **The discriminant** ●●● **[6 marks]** ☐ **Todo** ☐ **Mastered**

The equation $x^2 + (k+1)x + (k+3) = 0$, where $k$ is a constant, has no real roots.

(a) Show that $k^2 - 2k - 11 < 0$. **(3)**

(b) Hence find the set of possible values of $k$. **(3)**

WORKING

MY NOTES — 1.2 QUADRATICS: MISTAKES TO REMEMBER

# 1.3 Equations and Inequalities

Solve linear/quadratic simultaneous equations by substitution, and solve inequalities using a sketch so you pick the right region. Answers often need set notation.

## Key facts

- Quadratic inequality: find critical values, sketch, choose inside ($a < x < b$) or outside ($x < a$ or $x > b$)
- Set notation: $\{x : -2 < x < 5\}$, $\{x : x < 1\} \cup \{x : x > 4\}$
- Dashed lines for $<$ or $>$; solid lines for $\leq$ or $\geq$
- Multiplying or dividing an inequality by a negative reverses its sign

## Method

- Linear + quadratic: rearrange the linear equation for one variable and substitute into the quadratic.
- Always find both coordinates of every intersection point.
- For 'both inequalities hold', draw both solution sets on one number line and take the overlap.

## Exam traps

- Writing $2 < x > 5$ (meaningless).
- Using 'and' when you mean 'or' for outside regions.
- Dividing by a variable that could be negative.

**In the exam:** Typical 5-7 mark question: simultaneous equations followed by an inequality or region shading.

## Specification checklist

| Skill | Learnt | Practised | Mastered |
|---|---|---|---|
| Linear simultaneous equations | ☐ | ☐ | ☐ |
| Linear and quadratic simultaneous equations | ☐ | ☐ | ☐ |
| Linear inequalities | ☐ | ☐ | ☐ |
| Quadratic inequalities | ☐ | ☐ | ☐ |
| Inequalities on graphs and regions | ☐ | ☐ | ☐ |
| Set notation for solutions | ☐ | ☐ | ☐ |

## Videos (click to open)

☐ **Recap** — Equations & Inequalities in less than 20 minutes • A-Level Maths, Pure Year 1, Chapter 3 (Bicen Maths)
☐ Linear simultaneous equations — Simultaneous Equations: Examples of Using the Elimination Method (TLMaths)
☐ Linear and quadratic simultaneous equations — Simultaneous Equations: Examples of One Linear and One Quadratic Equation (TLMaths)
☐ Linear inequalities — Inequalities: Solving Linear Inequalities Graphically (TLMaths)
☐ Quadratic inequalities — Inequalities: Examples of Solving Quadratic Inequalities (TLMaths)
☐ Inequalities on graphs and regions — Inequalities: Draw and Shade a Region (TLMaths)
☐ Set notation for solutions — Inequalities: Introducing Inequalities, Set Notation and Interval Notation (TLMaths)

**Q1.3.01** Quadratic inequalities ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the set of values of $x$ for which $2x^2 - 6x - 20 < 0$. Give your answer in set notation.

**(3)**

WORKING

**Q1.3.02** Linear simultaneous equations ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve the simultaneous equations

$$-2x - 4y = 0$$

$$5x + y = -27$$

**(3)**

WORKING

**Q1.3.03** Linear inequalities ●○○ [2 marks] ☐ Todo ☐ Mastered

Find the set of values of $x$ for which $7x + 1 > 5x + 7$.

**(2)**

WORKING

**Q1.3.04** Quadratic inequalities ●○○ [3 marks] ☐ Todo ☐ Mastered

Find the set of values of $x$ for which $x^2 - 4x - 12 \leq 0$. Give your answer in set notation.

**(3)**

WORKING

**Q1.3.05** **Linear simultaneous equations** ●○○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Solve the simultaneous equations

$$5x + 4y = -23$$

$$2x + y = -8$$

**(3)**

WORKING

**Q1.3.06** **Linear inequalities** ●○○ **[2 marks]** ☐ **Todo** ☐ **Mastered**

Find the set of values of $x$ for which $2x - 9 > 1x + 9$.

**(2)**

WORKING

**Q1.3.07** **Quadratic inequalities** ●○○ **[3 marks]** ☐ **Todo** ☐ **Mastered**

Find the set of values of $x$ for which $x^2 - 8x + 15 \leq 0$. Give your answer in set notation.

**(3)**

WORKING

**Q1.3.08** Linear simultaneous equations ●○○ [3 marks] ☐ Todo ☐ Mastered

Solve the simultaneous equations

$$-4x - 5y = -42$$

$$-2x + 4y = 18$$

**(3)**

WORKING

**Q1.3.09** Linear inequalities ●○○ [2 marks] ☐ Todo ☐ Mastered

Find the set of values of $x$ for which $4x + 2 > 6x - 6$.

**(2)**

WORKING

**Q1.3.10** Set notation for solutions ●●○ [4 marks] ☐ Todo ☐ Mastered

Find the set of values of $x$ for which both $x - 8 < 0$ and $x^2 - 11x + 30 < 0$. Give your answer in set notation.

**(4)**

WORKING

**Q1.3.11** Inequalities on graphs and regions ●●○ [4 marks] ☐ Todo ☐ Mastered

On a sketch, shade the region $R$ satisfying $y > 2x^2 - 5$ and $y < 2x + 5$.
Use dashed or solid lines appropriately and find the $x$-coordinates of the points where the boundaries meet.

**(4)**

WORKING

**Q1.3.12** **Linear and quadratic simultaneous equations** ●●○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

Solve the simultaneous equations

$$y + x - 2 = 0$$

$$y = x^2 + 6x + 12$$

**(6)**

WORKING

**Q1.3.13** **Set notation for solutions** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

Find the set of values of $x$ for which both $x + 2 < 0$ and $x^2 + 3x < 0$. Give your answer in set notation.

**(4)**

WORKING

**Q1.3.14** Inequalities on graphs and regions ●●○ [4 marks] ☐ Todo ☐ Mastered

On a sketch, shade the region $R$ satisfying $y > x^2 - 7$ and $y < x + 7$.
Use dashed or solid lines appropriately and find the $x$-coordinates of the points where the boundaries meet.

**(4)**

WORKING

**Q1.3.15** Linear and quadratic simultaneous equations ●●○ [6 marks] ☐ Todo ☐ Mastered

Solve the simultaneous equations

$$y - 4x - 3 = 0$$

$$y = x^2 + x + 5$$

**(6)**

WORKING

**Q1.3.16** Set notation for solutions ●●○ [4 marks] ☐ Todo ☐ Mastered

Find the set of values of $x$ for which both $x - 4 < 0$ and $x^2 - 3x + 2 < 0$. Give your answer in set notation.

**(4)**

WORKING

**Q1.3.17** **Inequalities on graphs and regions** ●●○ **[4 marks]** ☐ **Todo** ☐ **Mastered**

On a sketch, shade the region $R$ satisfying $y > x^2 - 8$ and $y < -x + 8$.
Use dashed or solid lines appropriately and find the $x$-coordinates of the points where the boundaries meet.

**(4)**

WORKING

**Q1.3.18** **Linear and quadratic simultaneous equations** ●●○ **[6 marks]** ☐ **Todo** ☐ **Mastered**

Solve the simultaneous equations

$$y - 3x - 3 = 0$$

$$y = x^2 + 5x$$

**(6)**

WORKING

MY NOTES — 1.3 EQUATIONS AND INEQUALITIES: MISTAKES TO REMEMBER

# 1.4 Straight Line Graphs

Straight lines: gradient, equations in different forms, parallel and perpendicular lines, distances and midpoints, and using lines as simple models.

## Key facts

- $m = \frac{y_2 - y_1}{x_2 - x_1}$, $y - y_1 = m(x - x_1)$
- Parallel lines: $m_1 = m_2$; perpendicular lines: $m_1 m_2 = -1$
- Midpoint $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$, length $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- $ax + by + c = 0$ with $a, b, c$ integers is a common required form

## Method

- Perpendicular bisector: midpoint + negative reciprocal gradient.
- Area of a triangle formed with the axes: find both intercepts first.
- In modelling questions, interpret the gradient as a rate and the intercept as a starting value.

## Exam traps

- Leaving fractions when 'integers' are demanded.
- Swapping $x$ and $y$ differences in the gradient.

**In the exam:** Usually combined with circles or as part (a) of a longer coordinate geometry question.

## Specification checklist

| Skill | Learnt | Practised | Mastered |
|---|---|---|---|
| Gradient and y = mx + c | ☐ | ☐ | ☐ |
| y - y1 = m(x - x1) and ax + by + c = 0 | ☐ | ☐ | ☐ |
| Parallel and perpendicular lines | ☐ | ☐ | ☐ |
| Length and midpoint | ☐ | ☐ | ☐ |
| Modelling with straight lines | ☐ | ☐ | ☐ |

## Videos (click to open)

☐ **Recap** — Straight Line Graphs in less than 22 minutes • A-Level Maths, Pure Year 1, Chapter 5 (Bicen Maths)
☐ Gradient and y = mx + c — Coordinate Geometry: Finding the Equation of a Line in the form ax+by+c=0 (TLMaths)
☐ y - y1 = m(x - x1) and ax + by + c = 0 — Coordinate Geometry: The Equation of a Line in the form y-y1 = m(x - x1) (TLMaths)
☐ Parallel and perpendicular lines — Coordinate Geometry: Parallel and Perpendicular Lines (TLMaths)
☐ Length and midpoint — Coordinate Geometry: Examples of Finding the Midpoint between Two Points (TLMaths)
☐ Modelling with straight lines — Coordinate geometry (straight lines) : C1 Edexcel January 2011 Q9 (ExamSolutions)

**Q1.4.01** Parallel and perpendicular lines ●○○ [3 marks] ☐ Todo ☐ Mastered

The line $l_1$ has equation $y = -2x + 3$.
The line $l_2$ is perpendicular to $l_1$ and passes through $P(0, 5)$. Find an equation of $l_2$ in the form $y = mx + c$.

**(3)**

WORKING

**Q1.4.02** Length and midpoint ●○○ [3 marks] ☐ Todo ☐ Mastered

$A$ is the point $(-2, 7)$ and $B$ is the point $(-10, -8)$.

(a) Find the midpoint of $AB$. **(1)**

(b) Find the exact length of $AB$, simplifying your answer. **(2)**

WORKING

**Q1.4.03** Gradient and y = mx + c ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(5, 2)$ and $B(-5, -8)$ lie on the line $l$. Find an equation for $l$ in the form $ax + by + c = 0$, where $a$, $b$ and $c$ are integers.

**(4)**

WORKING

**Q1.4.04** Parallel and perpendicular lines ●○○ [3 marks] ☐ Todo ☐ Mastered

The line $l_1$ has equation $y = 1x + 7$.
The line $l_2$ is perpendicular to $l_1$ and passes through $P(-2, -3)$. Find an equation of $l_2$ in the form $y = mx + c$.

**(3)**

WORKING

**Q1.4.05** Length and midpoint ●○○ [3 marks] ☐ Todo ☐ Mastered

$A$ is the point $(-3, -2)$ and $B$ is the point $(-4, 5)$.

(a) Find the midpoint of $AB$. **(1)**

(b) Find the exact length of $AB$, simplifying your answer. **(2)**

WORKING

**Q1.4.06** Gradient and y = mx + c ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(0, -1)$ and $B(5, 2)$ lie on the line $l$. Find an equation for $l$ in the form $ax + by + c = 0$, where $a$, $b$ and $c$ are integers.

**(4)**

WORKING

**Q1.4.07** Parallel and perpendicular lines ●○○ [3 marks] ☐ Todo ☐ Mastered

The line $l_1$ has equation $y = 5x + 3$.

The line $l_2$ is parallel to $l_1$ and passes through $P(4, -3)$. Find an equation of $l_2$ in the form $y = mx + c$.

**(3)**

WORKING

**Q1.4.08** Length and midpoint ●○○ [3 marks] ☐ Todo ☐ Mastered

$A$ is the point $(4, 8)$ and $B$ is the point $(8, 12)$.

(a) Find the midpoint of $AB$. **(1)**

(b) Find the exact length of $AB$, simplifying your answer. **(2)**

WORKING

**Q1.4.09** Gradient and y = mx + c ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(-6, -1)$ and $B(-4, -4)$ lie on the line $l$. Find an equation for $l$ in the form $ax + by + c = 0$, where $a$, $b$ and $c$ are integers.

**(4)**

WORKING

**Q1.4.10** Modelling with straight lines ●●○ [6 marks] ☐ Todo ☐ Mastered

A plumber charges a fixed call-out fee plus an hourly rate. A job lasting 4 hours costs £154 and a job lasting 12 hours costs £346. The cost $C$ is modelled as a linear function of the time $t$.

(a) Find an equation linking $C$ and $t$. **(3)**

(b) Interpret the gradient and the intercept in context. **(2)**

(c) State a limitation of the model. **(1)**

WORKING

**Q1.4.11** y - y1 = m(x - x1) and ax + by + c = 0 ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the equation of the perpendicular bisector of the line segment joining $P(6, 12)$ and $Q(-12, -2)$. Give your answer in the form $ax + by + c = 0$, where $a$, $b$ and $c$ are integers.

**(5)**

WORKING

**Q1.4.12** Modelling with straight lines ●●○ [6 marks] ☐ Todo ☐ Mastered

A plumber charges a fixed call-out fee plus an hourly rate. A job lasting 5 hours costs £90 and a job lasting 12 hours costs £153. The cost $C$ is modelled as a linear function of the time $t$.

(a) Find an equation linking $C$ and $t$. **(3)**

(b) Interpret the gradient and the intercept in context. **(2)**

(c) State a limitation of the model. **(1)**

WORKING

**Q1.4.13** y - y1 = m(x - x1) and ax + by + c = 0 ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the equation of the perpendicular bisector of the line segment joining $P(4, 4)$ and $Q(8, -10)$. Give your answer in the form $ax + by + c = 0$, where $a$, $b$ and $c$ are integers.

**(5)**

WORKING

**Q1.4.14** Modelling with straight lines ●●○ [6 marks] ☐ Todo ☐ Mastered

The cost of hiring a van is a fixed fee plus a charge per day. A job lasting 4 days costs £68 and a job lasting 9 days costs £113. The cost $C$ is modelled as a linear function of the time $t$.

(a) Find an equation linking $C$ and $t$. **(3)**

(b) Interpret the gradient and the intercept in context. **(2)**

(c) State a limitation of the model. **(1)**

WORKING

**Q1.4.15** y - y1 = m(x - x1) and ax + by + c = 0 ●●○ [5 marks] ☐ Todo ☐ Mastered

Find the equation of the perpendicular bisector of the line segment joining $P(8, -10)$ and $Q(-2, 4)$. Give your answer in the form $ax + by + c = 0$, where $a$, $b$ and $c$ are integers.

**(5)**

WORKING

MY NOTES — 1.4 STRAIGHT LINE GRAPHS: MISTAKES TO REMEMBER

# 1.5 Circles

Circles: write the equation from centre and radius, complete the square to find centre and radius, and use circle properties (tangent perpendicular to radius, perpendicular bisector of a chord passes through the centre).

## Key facts

- $(x-a)^2+(y-b)^2=r^2$, centre $(a, b)$, radius $r$
- $x^2+y^2+2fx+2gy+c=0$ has centre $(-f, -g)$, $r=\sqrt{f^2+g^2-c}$
- Tangent at $P$ is perpendicular to the radius at $P$
- Perpendicular bisector of a chord passes through the centre; angle in a semicircle is $90^\circ$
- Line meets circle: substitute, then $b^2-4ac$ tells you secant ($>0$), tangent ($=0$), miss ($<0$)

## Method

- To find a circle through 3 points, intersect two perpendicular bisectors.
- For tangent lengths from an external point, use Pythagoras with the radius.
- To check if a point is inside, compare its distance from the centre with $r$.

## Exam traps

- Forgetting to square root $r^2$.
- Sign errors when completing the square: $(x-3)^2$ means centre $x=+3$.

**In the exam:** Paper 1 or 2, 6-10 marks: often centre/radius then a tangent or an intersection with a line.

## Specification checklist

| Skill | Learnt | Practised | Mastered |
|---|---|---|---|
| Equation of a circle | ☐ | ☐ | ☐ |
| Completing the square to find centre and radius | ☐ | ☐ | ☐ |
| Tangents and normals to circles | ☐ | ☐ | ☐ |
| Chords and perpendicular bisectors | ☐ | ☐ | ☐ |
| Intersections of lines and circles | ☐ | ☐ | ☐ |
| Circle theorems in coordinate geometry | ☐ | ☐ | ☐ |

## Videos (click to open)

☐ **Recap** — Circles in less than 22 minutes • A-Level Maths, Pure Year 1, Chapter 6 (Bicen Maths)
☐ Equation of a circle — Circles: Introducing the Equation of the Circle (TLMaths)
☐ Completing the square to find centre and radius — Circles: Using Completing the Square to find the Centre & Radius (TLMaths)
☐ Tangents and normals to circles — Circles: Finding Tangents & Normals (TLMaths)
☐ Chords and perpendicular bisectors — Circles: Perpendicular Bisector of Two Points on a Circle (TLMaths)
☐ Intersections of lines and circles — Circles: Finding where a Line Intersects a Circle (TLMaths)
☐ Circle theorems in coordinate geometry — Circles: The Angle in a Semicircle is a Right Angle (TLMaths)

**Q1.5.01** Completing the square to find centre and radius ●○○ [4 marks] ☐ Todo ☐ Mastered

The circle $C$ has equation $x^2 + y^2 + 4x - 8y - 30 = 0$. Find the coordinates of the centre of $C$ and the exact radius of $C$.

**(4)**

WORKING

**Q1.5.02** Equation of a circle ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(-5, 3)$ and $B(3, 5)$ are the ends of a diameter of the circle $C$. Find an equation for $C$.

**(4)**

WORKING

**Q1.5.03** Completing the square to find centre and radius ●○○ [4 marks] ☐ Todo ☐ Mastered

The circle $C$ has equation $x^2 + y^2 + 12x + 14y + 77 = 0$. Find the coordinates of the centre of $C$ and the exact radius of $C$.

**(4)**

WORKING

**Q1.5.04** Equation of a circle ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(5, 3)$ and $B(3, -1)$ are the ends of a diameter of the circle $C$. Find an equation for $C$.

**(4)**

WORKING

**Q1.5.05** Completing the square to find centre and radius ●○○ [4 marks] ☐ Todo ☐ Mastered

The circle $C$ has equation $x^2 + y^2 - 2x - 12y + 19 = 0$. Find the coordinates of the centre of $C$ and the exact radius of $C$.

**(4)**

WORKING

**Q1.5.06** Equation of a circle ●○○ [4 marks] ☐ Todo ☐ Mastered

The points $A(-2, 0)$ and $B(6, 8)$ are the ends of a diameter of the circle $C$. Find an equation for $C$.

**(4)**

WORKING

**Q1.5.07** Tangents and normals to circles ●●○ [6 marks] ☐ Todo ☐ Mastered

The circle $C$ has centre $(-2, 3)$ and passes through the point $P(0, 4)$.

(a) Find an equation for $C$. **(2)**

(b) Find an equation of the tangent to $C$ at $P$, giving your answer in the form $ax + by + c = 0$. **(4)**

WORKING

**Q1.5.08** Intersections of lines and circles ●●○ [5 marks] ☐ Todo ☐ Mastered

The circle $C$ has equation $(x+2)^2+(y-2)^2=5$ and the line $l$ has equation $y=-x+k$, where $k$ is a constant. Given that $l$ is a tangent to $C$, find the exact possible values of $k$.

**(5)**

WORKING

**Q1.5.09** Circle theorems in coordinate geometry ●●○ [6 marks] ☐ Todo ☐ Mastered

The points $A(-5,\ -1)$, $B(5,\ -1)$ and $C(3, 3)$ lie on a circle.

(a) Show that $AC$ is perpendicular to $BC$. **(3)**

(b) Hence explain why $AB$ is a diameter and find an equation of the circle. **(3)**

WORKING

**Q1.5.10** Tangents and normals to circles ●●○ [6 marks] ☐ Todo ☐ Mastered

The circle $C$ has centre $(-5, -4)$ and passes through the point $P(-2, -3)$.

(a) Find an equation for $C$. **(2)**

(b) Find an equation of the tangent to $C$ at $P$, giving your answer in the form $ax + by + c = 0$. **(4)**

WORKING

**Q1.5.11** Intersections of lines and circles ●●○ [5 marks] ☐ Todo ☐ Mastered

The circle $C$ has equation $(x-4)^2 + (y+4)^2 = 5$ and the line $l$ has equation $y = -x + k$, where $k$ is a constant. Given that $l$ is a tangent to $C$, find the exact possible values of $k$.

**(5)**

WORKING

**Q1.5.12** Circle theorems in coordinate geometry ●●○ [6 marks] ☐ Todo ☐ Mastered

The points $A(-8, 2)$, $B(2, 2)$ and $C(0, 6)$ lie on a circle.

(a) Show that $AC$ is perpendicular to $BC$. **(3)**

(b) Hence explain why $AB$ is a diameter and find an equation of the circle. **(3)**

WORKING

**Q1.5.13** Tangents and normals to circles ●●○ [6 marks] ☐ Todo ☐ Mastered

The circle $C$ has centre $(-4, 2)$ and passes through the point $P(-8, 5)$.

(a) Find an equation for $C$. **(2)**

(b) Find an equation of the tangent to $C$ at $P$, giving your answer in the form $ax + by + c = 0$. **(4)**

WORKING

**Q1.5.14** Intersections of lines and circles ●●○ [5 marks] ☐ Todo ☐ Mastered

The circle $C$ has equation $(x-4)^2 + (y-2)^2 = 10$ and the line $l$ has equation $y = -x + k$, where $k$ is a constant. Given that $l$ is a tangent to $C$, find the exact possible values of $k$.

**(5)**

WORKING

**Q1.5.15** Circle theorems in coordinate geometry ●●○ [6 marks] ☐ Todo ☐ Mastered

The points $A(-2, -3)$, $B(8, -3)$ and $C(6, 1)$ lie on a circle.

(a) Show that $AC$ is perpendicular to $BC$. **(3)**

(b) Hence explain why $AB$ is a diameter and find an equation of the circle. **(3)**

WORKING

**Q1.5.16** Chords and perpendicular bisectors ●●● [7 marks] ☐ Todo ☐ Mastered

The points $P(4, 1)$, $Q(-1, 6)$ and $R(-4, -3)$ lie on a circle $C$.

(a) Find the equations of the perpendicular bisectors of $PQ$ and $QR$. **(4)**

(b) Hence find the centre and radius of $C$. **(3)**

WORKING

**Q1.5.17** Chords and perpendicular bisectors ●●● [7 marks] ☐ Todo ☐ Mastered

The points $P(5, 2)$, $Q(-2, 1)$ and $R(6, -5)$ lie on a circle $C$.

(a) Find the equations of the perpendicular bisectors of $PQ$ and $QR$. **(4)**

(b) Hence find the centre and radius of $C$. **(3)**

WORKING

**Q1.5.18** **Chords and perpendicular bisectors** ●●● **[7 marks]** ☐ **Todo** ☐ **Mastered**

The points $P(3, 1)$, $Q(-7, 1)$ and $R(1, 5)$ lie on a circle $C$.

(a) Find the equations of the perpendicular bisectors of $PQ$ and $QR$. **(4)**

(b) Hence find the centre and radius of $C$. **(3)**

WORKING

MY NOTES — 1.5 CIRCLES: MISTAKES TO REMEMBER

# Answers and mark schemes

M = method mark, A = accuracy mark (depends on the preceding M), B = independent mark, ft = follow-through, * = given answer. Numerical answers are to 3 s.f. unless stated.

## 1.1 Algebraic Expressions

**Q1.1.01** **Expanding brackets · 3 marks**

$2x^3 - 3x^2 - 9x + 10$

M1 expands two brackets correctly · M1 multiplies by third bracket · A1 $2x^3 - 3x^2 - 9x + 10$

**Q1.1.02** **Surds · 2 marks**

$2\sqrt{6}$

M1 $\sqrt{96} = 4\sqrt{6}$ or $\sqrt{24} = 2\sqrt{6}$ · A1 $2\sqrt{6}$

**Q1.1.03** **Negative and fractional indices · 2 marks**

$3$

M1 $27^{1/3} = 3$ · A1 $3$

**Q1.1.04** **Factorising · 3 marks**

$2x(x-4)(x+7)$

M1 takes out $2x$ · M1 attempts to factorise the quadratic · A1 $2x(x-4)(x+7)$

**Q1.1.05** **Expanding brackets · 3 marks**

$x^3 + x^2 - 22x - 40$

M1 expands two brackets correctly · M1 multiplies by third bracket · A1 $x^3 + x^2 - 22x - 40$

**Q1.1.06** **Surds · 2 marks**

$-3\sqrt{6}$

M1 $\sqrt{54} = 3\sqrt{6}$ or $\sqrt{216} = 6\sqrt{6}$ · A1 $-3\sqrt{6}$

**Q1.1.07** **Negative and fractional indices · 2 marks**

$\frac{1}{7}$

M1 $49^{1/2} = 7$ · A1 $\frac{1}{7}$

**Q1.1.08** **Factorising · 3 marks**

$3x(3x-5)(3x+5)$

M1 takes out a common factor $3x$ · M1 difference of two squares · A1 $3x(3x-5)(3x+5)$

**Q1.1.09** **Expanding brackets · 3 marks**

$x^3 + 7x^2 + 4x - 12$

M1 expands two brackets correctly · M1 multiplies by third bracket · A1 $x^3 + 7x^2 + 4x - 12$

**Q1.1.10** **Surds · 2 marks**

$8\sqrt{3}$

M1 $\sqrt{27} = 3\sqrt{3}$ or $\sqrt{75} = 5\sqrt{3}$ · A1 $8\sqrt{3}$

**Q1.1.11** **Negative and fractional indices · 2 marks**

$4$

M1 $16^{1/2} = 4$ · A1 $4$

**Q1.1.12** **Index laws · 2 marks**

$\frac{64x^7}{3}$

M1 uses $(x^m)^n = x^{mn}$ to get $64x^{12}$ · A1 $\frac{64x^7}{3}$

**Q1.1.13** **Rationalising denominators · 3 marks**

$\frac{10}{19} - \frac{2\sqrt{6}}{19}$

M1 multiplies numerator and denominator by $5 - \sqrt{6}$ · M1 denominator $= 19$ · A1 $\frac{10}{19} - \frac{2\sqrt{6}}{19}$

**Q1.1.14** **Factorising · 3 marks**

$x(x+1)(x+6)$

M1 takes out $x$ · M1 attempts to factorise the quadratic · A1 $x(x+1)(x+6)$

**Q1.1.15** **Index laws · 3 marks**

$x = -\frac{8}{3}$

M1 writes every term as a power of 3 · M1 $2x + (x+4) = -4$ · A1 $x = -\frac{8}{3}$

**Q1.1.16** **Rationalising denominators · 4 marks**

$\frac{\sqrt{7}}{18} + \frac{23}{18}$

M1 multiplies by $3 - 3\sqrt{7}$ · M1 expands numerator (4 terms) · M1 denominator $-54$ · A1 $\frac{\sqrt{7}}{18} + \frac{23}{18}$

**Q1.1.17** **Index laws · 3 marks**

$x = \frac{5}{4}$

M1 writes every term as a power of 2 · M1 $3x + (x+1) = 6$ · A1 $x = \frac{5}{4}$

**Q1.1.18** **Rationalising denominators · 4 marks**

$\frac{11}{29} - \frac{\sqrt{5}}{29}$

M1 multiplies by $4 - 3\sqrt{5}$ · M1 expands numerator (4 terms) · M1 denominator $-29$ · A1 $\frac{11}{29} - \frac{\sqrt{5}}{29}$

## 1.2 Quadratics

**Q1.2.01** **Quadratic functions and graphs · 6 marks**

**(a)** $(-3, 0), (3, 0), (0, -9)$

M1 factorises or solves $y = 0$ · A1 both $x$-intercepts · B1 $y$-intercept

**(b)** $(0, -9)$, minimum

M1 symmetry or completing the square for $x$ · A1 coordinates · B1 nature

**Q1.2.02** **Quadratic formula · 3 marks**

$x = \frac{1 \pm \sqrt{19}}{2}$

M1 correct substitution into the formula · M1 discriminant $= 76$ · A1 both roots, simplified

**Q1.2.03** **Solving by factorising · 3 marks**

$x = -3,\ x = 5$

M1 factorises: $(2x + 6)(x - 5)$ · A1 one correct root · A1 both roots

**Q1.2.04** **Quadratic functions and graphs · 6 marks**

**(a)** $(-5, 0), (5, 0), (0, 25)$

M1 factorises or solves $y = 0$ · A1 both $x$-intercepts · B1 $y$-intercept

**(b)** $(0, 25)$, maximum

M1 symmetry or completing the square for $x$ · A1 coordinates · B1 nature

**Q1.2.05** **Quadratic formula · 3 marks**

$x = 1 \pm 2\sqrt{2}$

M1 correct substitution into the formula · M1 discriminant $= 32$ · A1 both roots, simplified

**Q1.2.06** **Solving by factorising · 3 marks**

$x = -2,\ x = -1$

M1 factorises: $(2x + 2)(x + 2)$ · A1 one correct root · A1 both roots

**Q1.2.07** **Completing the square · 3 marks**

**(a)** $(x - \frac{1}{2})^2 - \frac{1}{4}$

B1 $p = -\frac{1}{2}$ · M1A1 $q = -\frac{1}{4}$

**(b)** $(\frac{1}{2}, -\frac{1}{4})$

B1ft

**Q1.2.08** **Quadratic functions and graphs · 6 marks**

**(a)** $(4, 0), (5, 0), (0, -20)$

M1 factorises or solves $y = 0$ · A1 both $x$-intercepts · B1 $y$-intercept

**(b)** $(\frac{9}{2}, \frac{1}{4})$, maximum

M1 symmetry or completing the square for $x$ · A1 coordinates · B1 nature

**Q1.2.09** **Quadratic formula · 3 marks**

$x = \frac{-3 \pm \sqrt{30}}{3}$

M1 correct substitution into the formula · M1 discriminant $= 120$ · A1 both roots, simplified

**Q1.2.10** **Solving by factorising · 3 marks**

$x = -5,\ x = 2$

M1 factorises: $(x - 2)(x + 5)$ · A1 one correct root · A1 both roots

**Q1.2.11** **Completing the square · 4 marks**

$-2(x + 1)^2 + 3$; maximum value 3

B1 $a = -2$ · M1 $b = 1$ · A1 $c = 3$ · B1ft max value $= 3$

**Q1.2.12** **The discriminant · 3 marks**

$k > \frac{9}{4}$

M1 uses $b^2 - 4ac$ with $a = 1,\ b = 3,\ c = k$ · M1 correct inequality $< 0$ · A1 $k > \frac{9}{4}$

**Q1.2.13** **Hidden quadratics · 4 marks**

$x = 1,\ x = 16$

M1 lets $u = \sqrt{x}$ · M1 $(u - 1)(u - 4) = 0$ · M1 squares to find $x$ · A1 both values

**Q1.2.14** **Completing the square · 4 marks**

$-1(x+3)^2+0$; maximum value $0$

B1 $a=-1$ · M1 $b=3$ · A1 $c=0$ · B1ft max value $=0$

**Q1.2.15** **The discriminant · 3 marks**

$k<\frac{9}{2}$

M1 uses $b^2-4ac$ with $a=2$, $b=-6$, $c=k$ · M1 correct inequality $>0$ · A1 $k<\frac{9}{2}$

**Q1.2.16** **Hidden quadratics · 4 marks**

$x=\pm1$, $x=\pm2$

M1 lets $u=x^2$ · M1 $(u-1)(u-4)=0$ · M1 square roots · A1 all four

**Q1.2.17** **Hidden quadratics · 4 marks**

$x=\pm1$, $x=\pm4$

M1 lets $u=x^2$ · M1 $(u-1)(u-16)=0$ · M1 square roots · A1 all four

**Q1.2.18** **The discriminant · 6 marks**

**(a)** $k^2-2k-11<0$

M1 uses $b^2-4ac<0$ · M1 expands · A1* given answer, no errors

**(b)** $1-2\sqrt{3}<k<1+2\sqrt{3}$

M1 solves the quadratic to find critical values · M1 chooses the inside region · A1 $1-2\sqrt{3}<k<1+2\sqrt{3}$

## 1.3 Equations and Inequalities

**Q1.3.01** **Quadratic inequalities · 3 marks**

$-2<x<5$, i.e. $\{x:-2<x<5\}$

M1 critical values $x=-2,5$ · M1 chooses inside region · A1 correct set notation

**Q1.3.02** **Linear simultaneous equations · 3 marks**

$x=-6$, $y=3$

M1 eliminates one variable · A1 one value · A1 both

**Q1.3.03** **Linear inequalities · 2 marks**

$x>3$

M1 collects terms · A1 $x>3$

**Q1.3.04** **Quadratic inequalities · 3 marks**

$-2\le x\le 6$, i.e. $\{x:-2\le x\le 6\}$

M1 critical values $x=-2,6$ · M1 chooses inside region · A1 correct set notation

**Q1.3.05** **Linear simultaneous equations · 3 marks**

$x=-3$, $y=-2$

M1 eliminates one variable · A1 one value · A1 both

**Q1.3.06** **Linear inequalities · 2 marks**

$x>18$

M1 collects terms · A1 $x>18$

**Q1.3.07** **Quadratic inequalities · 3 marks**

$3\le x\le 5$, i.e. $\{x:3\le x\le 5\}$

M1 critical values $x=3,5$ · M1 chooses inside region · A1 correct set notation

**Q1.3.08** **Linear simultaneous equations · 3 marks**

$x=3$, $y=6$

M1 eliminates one variable · A1 one value · A1 both

**Q1.3.09** **Linear inequalities · 2 marks**

$x>4$

M1 collects terms · A1 $x>4$

**Q1.3.10** **Set notation for solutions · 4 marks**

$\{x:5<x<6\}$

B1 $x<8$ · M1A1 $5<x<6$ · A1 combined set

**Q1.3.11** **Inequalities on graphs and regions · 4 marks**

Boundaries are dashed (strict). Intersections where $2x^2-2x-10=0$: $x=\frac{1}{2}-\frac{\sqrt{21}}{2}$, $x=\frac{1}{2}+\frac{\sqrt{21}}{2}$

B1 parabola and line sketched · B1 dashed lines · M1 solves for intersections · A1 region R shaded between the curves

**Q1.3.12** **Linear and quadratic simultaneous equations · 6 marks**

$x=-5, y=7$; $x=-2, y=4$

M1 substitutes linear into quadratic · A1 $x^2+7x+10=0$ · M1 solves 3TQ · A1 both $x$ values · M1 substitutes back · A1 both pairs

**Q1.3.13** **Set notation for solutions · 4 marks**

$\{x : -3 < x < -2\}$

B1 $x < -2$ · M1A1 $-3 < x < 0$ · A1 combined set

**Q1.3.14** **Inequalities on graphs and regions · 4 marks**

Boundaries are dashed (strict). Intersections where $x^2 - x - 14 = 0$: $x = \frac{1}{2} - \frac{\sqrt{57}}{2}$, $x = \frac{1}{2} + \frac{\sqrt{57}}{2}$

B1 parabola and line sketched · B1 dashed lines · M1 solves for intersections · A1 region R shaded between the curves

**Q1.3.15** **Linear and quadratic simultaneous equations · 6 marks**

$x = 1, y = 7; x = 2, y = 11$

M1 substitutes linear into quadratic · A1 $x^2 - 3x + 2 = 0$ · M1 solves 3TQ · A1 both $x$ values · M1 substitutes back · A1 both pairs

**Q1.3.16** **Set notation for solutions · 4 marks**

$\{x : 1 < x < 2\}$

B1 $x < 4$ · M1A1 $1 < x < 2$ · A1 combined set

**Q1.3.17** **Inequalities on graphs and regions · 4 marks**

Boundaries are dashed (strict). Intersections where $x^2 + x - 16 = 0$: $x = -\frac{1}{2} + \frac{\sqrt{65}}{2}$, $x = -\frac{\sqrt{65}}{2} - \frac{1}{2}$

B1 parabola and line sketched · B1 dashed lines · M1 solves for intersections · A1 region R shaded between the curves

**Q1.3.18** **Linear and quadratic simultaneous equations · 6 marks**

$x = -3, y = -6; x = 1, y = 6$

M1 substitutes linear into quadratic · A1 $x^2 + 2x - 3 = 0$ · M1 solves 3TQ · A1 both $x$ values · M1 substitutes back · A1 both pairs

## 1.4 Straight Line Graphs

**Q1.4.01** **Parallel and perpendicular lines · 3 marks**

$y = \frac{1}{2}x + 5$

B1 gradient $\frac{1}{2}$ · M1 uses the point · A1 $y = \frac{1}{2}x + 5$

**Q1.4.02** **Length and midpoint · 3 marks**

**(a)** $(-6, -\frac{1}{2})$

B1

**(b)** $17$

M1 $\sqrt{8^2 + 15^2}$ · A1 $17$

**Q1.4.03** **Gradient and y = mx + c · 4 marks**

$x - y - 3 = 0$

M1 gradient $= 1$ · M1 uses $y - y_1 = m(x - x_1)$ · A1 any correct form · A1 integer form

**Q1.4.04** **Parallel and perpendicular lines · 3 marks**

$y = -1x - 5$

B1 gradient $-1$ · M1 uses the point · A1 $y = -1x - 5$

**Q1.4.05** **Length and midpoint · 3 marks**

**(a)** $(-\frac{7}{2}, \frac{3}{2})$

B1

**(b)** $5\sqrt{2}$

M1 $\sqrt{1^2 + 7^2}$ · A1 $5\sqrt{2}$

**Q1.4.06** **Gradient and y = mx + c · 4 marks**

$3x - 5y - 5 = 0$

M1 gradient $= \frac{3}{5}$ · M1 uses $y - y_1 = m(x - x_1)$ · A1 any correct form · A1 integer form

**Q1.4.07** **Parallel and perpendicular lines · 3 marks**

$y = 5x - 23$

B1 gradient $5$ · M1 uses the point · A1 $y = 5x - 23$

**Q1.4.08** **Length and midpoint · 3 marks**

**(a)** $(6, 10)$

B1

**(b)** $4\sqrt{2}$

M1 $\sqrt{4^2 + 4^2}$ · A1 $4\sqrt{2}$

**Q1.4.09** **Gradient and y = mx + c · 4 marks**

$3x + 2y + 20 = 0$

M1 gradient $= -\frac{3}{2}$ · M1 uses $y - y_1 = m(x - x_1)$ · A1 any correct form · A1 integer form

**Q1.4.10** **Modelling with straight lines · 6 marks**

**(a)** $C = 24t + 58$

M1 gradient $= 24$ · M1 substitutes a point · A1 $C = 24t + 58$

**(b)** £24 per hour; fixed charge £58

B1 gradient · B1 intercept

**(c)** e.g. charges may be per part-hour / discounts for long jobs

B1 sensible limitation

**Q1.4.11** **y - y1 = m(x - x1) and ax + by + c = 0 · 5 marks**

$9x + 7y - 8 = 0$

B1 midpoint $(-3, 5)$ · M1 gradient of $PQ$ · M1 perpendicular gradient $-\frac{9}{7}$ · M1 line through midpoint · A1 integer form

**Q1.4.12** **Modelling with straight lines · 6 marks**

**(a)** $C = 9t + 45$

M1 gradient $= 9$ · M1 substitutes a point · A1 $C = 9t + 45$

**(b)** £9 per hour; fixed charge £45

B1 gradient · B1 intercept

**(c)** e.g. charges may be per part-hour / discounts for long jobs

B1 sensible limitation

**Q1.4.13** **y - y1 = m(x - x1) and ax + by + c = 0 · 5 marks**

$2x - 7y - 33 = 0$

B1 midpoint $(6, -3)$ · M1 gradient of $PQ$ · M1 perpendicular gradient $\frac{2}{7}$ · M1 line through midpoint · A1 integer form

**Q1.4.14** **Modelling with straight lines · 6 marks**

**(a)** $C = 9t + 32$

M1 gradient $= 9$ · M1 substitutes a point · A1 $C = 9t + 32$

**(b)** £9 per day; fixed charge £32

B1 gradient · B1 intercept

**(c)** e.g. charges may be per part-hour / discounts for long jobs

B1 sensible limitation

**Q1.4.15** **y - y1 = m(x - x1) and ax + by + c = 0 · 5 marks**

$5x - 7y - 36 = 0$

B1 midpoint $(3, -3)$ · M1 gradient of $PQ$ · M1 perpendicular gradient $\frac{5}{7}$ · M1 line through midpoint · A1 integer form

## 1.5 Circles

**Q1.5.01** **Completing the square to find centre and radius · 4 marks**

centre $(-2, 4)$, radius $5\sqrt{2}$

M1 completes the square in $x$ and $y$ · A1 centre · M1 $r^2 = 4 + 16 - (-30)$ · A1 $r = 5\sqrt{2}$

**Q1.5.02** **Equation of a circle · 4 marks**

$(x + 1)^2 + (y - 4)^2 = 17$

M1A1 centre $(-1, 4)$ · M1 $r^2 = 17$ · A1 equation

**Q1.5.03** **Completing the square to find centre and radius · 4 marks**

centre $(-6, -7)$, radius $2\sqrt{2}$

M1 completes the square in $x$ and $y$ · A1 centre · M1 $r^2 = 36 + 49 - (77)$ · A1 $r = 2\sqrt{2}$

**Q1.5.04** **Equation of a circle · 4 marks**

$(x - 4)^2 + (y - 1)^2 = 5$

M1A1 centre $(4, 1)$ · M1 $r^2 = 5$ · A1 equation

**Q1.5.05** **Completing the square to find centre and radius · 4 marks**

centre $(1, 6)$, radius $3\sqrt{2}$

M1 completes the square in $x$ and $y$ · A1 centre · M1 $r^2 = 1 + 36 - (19)$ · A1 $r = 3\sqrt{2}$

**Q1.5.06** **Equation of a circle · 4 marks**

$(x - 2)^2 + (y - 4)^2 = 32$

M1A1 centre $(2, 4)$ · M1 $r^2 = 32$ · A1 equation

**Q1.5.07** **Tangents and normals to circles · 6 marks**

**(a)** $(x + 2)^2 + (y - 3)^2 = 5$

M1 radius$^2$ by Pythagoras · A1

**(b)** $2x + y - 4 = 0$

M1 gradient of radius $\frac{1}{2}$ · M1 tangent gradient $-2$ · M1 line through $P$ · A1

**Q1.5.08** **Intersections of lines and circles · 5 marks**

$k = -\sqrt{10}$ or $k = \sqrt{10}$

M1 substitutes the line into the circle · A1 correct 3TQ in $x$ · M1 uses $b^2 - 4ac = 0$ · M1 solves for $k$ · A1 both values

**Q1.5.09** **Circle theorems in coordinate geometry · 6 marks**

**(a)** gradients $\frac{1}{2}$ and $-2$; product $-1$

M1 gradient of $AC$ · M1 gradient of $BC$ · A1 product $=-1$ so perpendicular

**(b)** angle in a semicircle is $90°$; $x^2+(y+1)^2=25$

B1 angle in semicircle · M1 centre = midpoint of $AB$ · A1 equation

**Q1.5.10** **Tangents and normals to circles · 6 marks**

**(a)** $(x+5)^2+(y+4)^2=10$

M1 radius² by Pythagoras · A1

**(b)** $3x+y+9=0$

M1 gradient of radius $\frac{1}{3}$ · M1 tangent gradient $-3$ · M1 line through $P$ · A1

**Q1.5.11** **Intersections of lines and circles · 5 marks**

$k=-\sqrt{10}$ or $k=\sqrt{10}$

M1 substitutes the line into the circle · A1 correct 3TQ in $x$ · M1 uses $b^2-4ac=0$ · M1 solves for $k$ · A1 both values

**Q1.5.12** **Circle theorems in coordinate geometry · 6 marks**

**(a)** gradients $\frac{1}{2}$ and $-2$; product $-1$

M1 gradient of $AC$ · M1 gradient of $BC$ · A1 product $=-1$ so perpendicular

**(b)** angle in a semicircle is $90°$; $(x+3)^2+(y-2)^2=25$

B1 angle in semicircle · M1 centre = midpoint of $AB$ · A1 equation

**Q1.5.13** **Tangents and normals to circles · 6 marks**

**(a)** $(x+4)^2+(y-2)^2=25$

M1 radius² by Pythagoras · A1

**(b)** $4x-3y+47=0$

M1 gradient of radius $-\frac{3}{4}$ · M1 tangent gradient $\frac{4}{3}$ · M1 line through $P$ · A1

**Q1.5.14** **Intersections of lines and circles · 5 marks**

$k=6-2\sqrt{5}$ or $k=2\sqrt{5}+6$

M1 substitutes the line into the circle · A1 correct 3TQ in $x$ · M1 uses $b^2-4ac=0$ · M1 solves for $k$ · A1 both values

**Q1.5.15** **Circle theorems in coordinate geometry · 6 marks**

**(a)** gradients $\frac{1}{2}$ and $-2$; product $-1$

M1 gradient of $AC$ · M1 gradient of $BC$ · A1 product $=-1$ so perpendicular

**(b)** angle in a semicircle is $90°$; $(x-3)^2+(y+3)^2=25$

B1 angle in semicircle · M1 centre = midpoint of $AB$ · A1 equation

**Q1.5.16** **Chords and perpendicular bisectors · 7 marks**

**(a)** two lines through the midpoints, perpendicular to $PQ$ and $QR$

M1A1 first bisector · M1A1 second bisector

**(b)** centre $(-1, 1)$, radius $5$

M1 solves simultaneously · A1 centre · B1 radius

**Q1.5.17** **Chords and perpendicular bisectors · 7 marks**

**(a)** two lines through the midpoints, perpendicular to $PQ$ and $QR$

M1A1 first bisector · M1A1 second bisector

**(b)** centre $(2,\ -2)$, radius $5$

M1 solves simultaneously · A1 centre · B1 radius

**Q1.5.18** **Chords and perpendicular bisectors · 7 marks**

**(a)** two lines through the midpoints, perpendicular to $PQ$ and $QR$

M1A1 first bisector · M1A1 second bisector

**(b)** centre $(-2, 1)$, radius $5$

M1 solves simultaneously · A1 centre · B1 radius